

Chapter 6: In-Class Exercises

Part 1

6.2 Problem: An eight-year \$3,000 10% bond with semiannual coupons and redemption value \$2,800 is bought at a price to give the investor an annual effective yield rate of 12%. Find the base amount and the price of the bond.

$$N = 8$$

$$n = 16$$

$$F = 3000$$

$$\rightarrow j = (1.12)^{1/2} - 1 = .0583$$

$$\alpha = 10\%$$

$$r = 5\%$$

$$m = 2$$

$$C = 2800$$

$$P = 3000(.05) a_{\overline{16}|.0583} + 2800(1.0583)^{-16}$$
$$= 2664.61$$

$$i = 12\%$$

$$G = 150/.0583 = 2572.88$$

6.2 Problem: A twelve-year \$2,000 8% par-value bond with quarterly coupons is bought for \$2,200. Find the effective yield rate per coupon period j , the annual effective yield i , and the nominal yield rate I convertible quarterly.

$$N = 12$$

$$n = 48$$

$$F = C = 2000$$

$$r = .02$$

$$\alpha = 8\%$$

$$m = 4$$

$$P = 2200$$

calculator

$$N = 48$$

$$PMT = 2000(.02) = 40$$

$$PV = -2200$$

$$FV = 2000$$

$$I/Y = 1.69394933\% = j$$

$$I = 6.77579730\%$$

$$i = 6.9499177\%$$

6.2

Problem: A nine-year \$5,000 7% bond with semiannual coupons is purchased for \$4,986. This price allows the purchaser an annual effective yield of 6%. Calculate the redemption amount.

$$N = 9$$

$$n = 18$$

$$F = 5000$$

$$r = 3.5\%$$

$$\alpha = 7\% \rightarrow j = 1.06^{1/2} - 1$$

$$m = 2$$

$$= .029563014$$

$$P = 4986$$

$$i = 6\%$$

$$4986 = 5000(.035) a_{\overline{18}|j} + C(1+j)^{-18}$$

$$C = 4342.33$$

6.2

Problem: A twenty-year \$1,000 bond with semiannual coupons is redeemable at par and has a nominal coupon rate of 8% for the first five years, 9% for the next five years, and 10% for the final ten years. Amy Astacio purchases the bond so as to receive a nominal yield of 9.2%. How much does she pay?

$$N = 20$$

$$n = 40$$

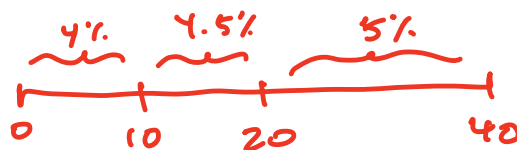
$$F = 1000 = C$$

$$r_1 = 4\%$$

$$m = 2$$

$$r_2 = 4.5\%$$

$$r_3 = 5\%$$



$$\alpha_1 = 8\% \leftarrow \text{first } 5 \text{ yrs}$$

$$\alpha_2 = 9\% \leftarrow \text{next } 5 \text{ yrs}$$

$$\alpha_3 = 10\% \leftarrow \text{final } 10 \text{ yrs}$$

$$I = 9.2\% \quad j = .046$$

$$P = 40 a_{\overline{10}|.046} + 45 a_{\overline{10}|.046} (1.046)^{-10} + 50 a_{\overline{20}|.046} (1.046)^{-20} + 1000 (1.046)^{-40} = 968.72$$

6.2

Problem: Sydney and Sons Tool Corporation has structured its bond offering to appeal to the inflation conscious investor. This is done projecting an annual rate of inflation of 2.5%. An eight-year \$1,000 face value bond with annual coupons has a redemption amount equal to $\$1,218.40 \approx \$1,000(1.025)^8$, an initial coupon of \$80, and coupons that increase by 2.5% annually. How much should an investor who wishes to earn a noninflation-adjusted rate of 8% be willing to pay for the bond? If the actual rate of inflation is 2%, what is the investor's inflation-adjusted yield?

$$\begin{aligned}
 N &= 8 \\
 F &= 1000 \\
 m &= 1 \\
 C &= 1218.40 \\
 j &= 8\% \\
 j &= 2\%
 \end{aligned}$$

coupons = 80, 80(1.025), 80(1.025)², ...

$$P = 80 \left(\frac{1 - \left(\frac{1.025}{1.08} \right)^8}{.08 - .025} \right) + 1218.40(1.08)^{-8}$$

$$= 1155.33$$

$$P = 80 \left(\frac{1 - \left(\frac{1.025}{1.02} \right)^8}{.02 - .025} \right) + 1218.40(1.02)^{-8} = 1678.21$$

Problem: A bond with a face value of \$6,000 and an annual coupon rate of 12% convertible semiannually will mature in ten years for its face value. If the bond is priced using a nominal yield rate of 6% convertible semiannually, what is the amount of premium or discount in this bond?

6.3

$$\begin{aligned}
 N &= 10 \\
 F &= 6000 \\
 C &= 6000 \\
 \alpha &= 12\% \\
 m &= 2 \\
 I &= 6\% \\
 j &= 3\%
 \end{aligned}$$

$$\begin{aligned}
 n &= 20 \\
 r &= 6\% \\
 Fr &= Cg \\
 \Downarrow \\
 g &= 6\%
 \end{aligned}$$

$$P - C = C(g - j) a_{\overline{n}|j}$$

premium since $g > j$

$$\begin{aligned}
 P - C &= 6000(.06 - .03) a_{\overline{20}|.03} \\
 &= 2677.95
 \end{aligned}$$

6.3

Problem: A $7\frac{1}{2}$ -year 14% bond with a face value of \$2,500 has semiannual coupons and is sold to yield 7.2% convertible semiannually. The discount on the bond is \$283.12. Find the price of the bond.

$$N = 7.5$$

$$\alpha = 14\%$$

$$F = 2500$$

$$m = 2$$

$$I = 7.2\%$$

$$n = 15$$

$$r = 7\%$$

$$j = .036$$

$$\text{discount} = 283.12 = C - P = C(.036 - g)a_{\overline{15}|.036}$$

$$Fr = Cg \text{ so } g = \frac{175}{C}$$

$$283.12 = C\left(.036 - \frac{175}{C}\right)a_{\overline{15}|.036}$$

$$\Rightarrow C = 5548.81$$

$$\Rightarrow P = 5265.69$$

6.4

Problem: A \$2,000 bond with coupon rate of 10% payable quarterly is redeemable after an unspecified number of periods for \$2,250. The bond is bought to yield 8% convertible quarterly. If the present value of the redemption is \$869.71, find the purchase price.

$$F = 2000$$

$$\alpha = 10\%$$

$$m = 4$$

$$C = 2250$$

$$I = 8\%$$

$$K = 869.71$$

$$n = ?$$

$$r = 2.5\%$$

$$j = 2\%$$

$$P = Fra_{\overline{n}|j} + K$$

$$P = 2000(.025)a_{\overline{n}|.02} + 869.71$$

$$869.71 = 2250(1+j)^{-n} \text{ or } 869.71(1.02)^n = 2250$$

$$n = \ln\left(\frac{2250}{869.71}\right) / \ln(1.02) = 48$$

$$P = 2403.37$$

6.4

Problem: An eight-year bond has annual coupons and a redemption value of \$2,338. It is purchased to yield 9% and each coupon is for \$63. Use the base amount formula to calculate the price of the bond.

$$N = 8$$

$$m = 1$$

$$C = 2338$$

$$j = 9\%$$

$$\text{Coupon} = 63$$

$$P = 63 a_{\overline{8}|.09} + 2338 (1.09)^{-8}$$

$$= 1522.06$$

$$G_j = 63 \Rightarrow G = 700$$

$$(C - G) v_j^n + G = (2338 - 700) (1.09)^{-8} + 700$$

$$= 1522.06$$

TYPE	LETTER	MEANING
extent	N	number of years in the bond term
frequency	m	number of coupons per year
extent	n	number of coupons, $n = Nm$
Coupon rate	α	nominal coupon rate, annual coupons total $F\alpha$
Coupon rate	r	coupon rate per coupon period, amount of each coupon is Fr
Coupon rate	g	modified coupon rate, coupon amount is Cg
Yield rate	I	nominal yield rate convertible m times per year
Yield rate	j	effective yield rate for the coupon period
Yield rate	i	annual effective yield rate
Dollar amount	F	face (or par) value, coupon amount is Fr
Dollar amount	C	redemption amount
Dollar amount	G	base amount, coupon amount is Gj
Dollar amount	P	price at issue
Dollar amount	K	present value of the redemption amount

TABLE (6.2.4) Bond variables

6.5

Problem: A six-year \$1,800 8.5% bond with semiannual coupons is bought for \$1,918 and is redeemable for \$1,860. Find the amount for amortization of premium at the end of two-and-a-half years and the interest due at that time.

$$N = 6$$

$$n = 12$$

$$1918 = 76.5 a_{\overline{12}|j} + 1860(1+j)^{-12}$$

$$F = 1800$$

$$r = 4.25\%$$

$$j = 3.78477965$$

$$\alpha = 8.5\% \rightarrow$$

$$m = 2$$

$$P = 1918$$

$$Pr(5) = PMT - In(5)$$

$$= 76.5 - B(4)j$$

$$= 76.5 - j(1918(1+j)^4 - 76.5 s_{\overline{4}|j})$$

$$= 4.53$$

$$In(5) = 71.97$$

$$C = 1860$$

6.5

Problem: A twelve-year bond with semiannual level coupons is bought at a premium to yield 7.5% convertible semiannually. If the amount for amortization of premium in the fourth to the last payment is \$8.02, find the premium and the total amount for amortization of premium in the first three years.

$$N = 12 \rightarrow n = 24$$

$$m = 2$$

$$I = 7.5\% \rightarrow j = .0375$$

$$\text{Find } \sum_{i=1}^6 Pr(i)$$

$$Pr(21) = 8.02$$

$$= B(0) - B(6)$$

$$Pr(21) = 8.02 \Rightarrow 8.02 = C(g - .0375)(1.0375)^{-4}$$

$$9.292376 = C(g - .0375)$$

$$B(6) = C a_{\overline{18}|.0375} (g - .0375) + C$$

$$B(0) = C a_{\overline{24}|.0375} (g - .0375) + C$$

$$B(0) - B(6) = C(g - .0375)(a_{\overline{24}|.0375} - a_{\overline{18}|.0375}) = 25.32$$

6.5

Problem: A \$10,000 par-value twenty-year 14% bond with annual coupons is bought for \$9,562. Find the amount for accumulation of discount in the third coupon and the amount of interest for the third year.

$$F = 10,000 = C$$

$$N = 20 \rightarrow n = 20$$

$$\alpha = 14\% \rightarrow r = 14\%$$

$$m = 1$$

$$P = 9562$$

calculator

$$N = 20$$

$$PMT = 1400$$

$$PV = -9562$$

$$FV = 10,000$$

$$\rightarrow j = 14.68768719$$

Find $Pr(3)$ and $In(3)$

$$Pmt = Fr = 1400$$

$$In(3) = jB(2)$$

$$= j(9562(1+j)^2 - 1400s_{\overline{2}|j})$$

$$= 1405.84$$

$$Pr(3) = 1405.85 - 1400 = 5.85$$

6.5

Problem: An n -year \$1,000 par-value bond with 8% annual coupons has an annual effective yield of i , $1 + i > 0$. The book value of the bond at the end of the third year is \$990.92 and the book value of the bond at the end of the fifth year is \$995.26. Find the price of the bond.

$$F = 1000 = C$$

$$m = 1$$

$$\alpha = r = 8\%$$

$$B(3) = 990.92$$

$$B(5) = 995.26$$

Find P

$$B(t+1) = B(t)(1+j) - Fr$$

$$B(4) = B(3)(1+j) - 80$$

$$B(5) = B(4)(1+j) - 80$$

$$= (B(3)(1+j) - 80)(1+j) - 80$$

$$995.26 = 990.92(1+j)^2 - 80(1+j) - 80$$

$$j = .082835847$$

$$B(3) = 80a_{\overline{n-3}|j} + 1000(1+j)^{-(n-3)} \Rightarrow n = 6.87262614$$

$$P = 985.58$$

6.5

Problem: An 8% par-value bond with face amount \$100,000 and semiannual coupons matures after four years. Construct the amortization schedule over its term if the investor's nominal yield rate is 6% convertible semiannually.

$$\begin{aligned}
 F &= 100,000 \\
 C &= 100,000 \\
 \alpha &= 8\% \\
 m &= 2 \\
 N &= 4 \\
 I &= 6\% \\
 n &= 8 \\
 r &= 4\% \\
 j &= 3\%
 \end{aligned}$$

t	PMT	$In(t)$	$Pr(t)$	$B(t)$
0	—	—	—	107,019.69
1	4000	3210.59	789.41	106,230.28
2	4000	3186.91	813.09	105,417.19
3	4000	3162.52	837.48	104,579.71
4	4000	3137.39	862.61	103,717.10
5	4000	3111.51	888.49	102,828.61
6	4000	3084.86	915.14	101,913.47
7	4000	3057.40	942.60	100,970.87
8	104,000	3029.13	100,970.87	0

6.5

Problem: An 8% par-value bond with face amount \$100,000 and semiannual coupons matures after four years. Construct the amortization schedule over its term if the investor's nominal yield rate is 10% convertible semiannually.

$$\begin{aligned}
 F &= 100,000 \\
 C &= 100,000 \\
 \alpha &= 8\% \\
 m &= 2 \\
 N &= 4 \\
 I &= 10\% \\
 n &= 8 \\
 r &= 4\% \\
 j &= 5\%
 \end{aligned}$$

t	PMT	$In(t)$	$Pr(t)$	$B(t)$
0	—	—	—	93,536.79
1	4000	4676.84	-676.84	94,213.63
2	4000	4710.68	-710.68	94,924.31
3	4000	4746.22	-746.22	95,670.53
4	4000	4783.53	-783.53	96,454.05
5	4000	4822.70	-822.70	97,276.76
6	4000	4863.84	-863.84	98,140.59
7	4000	4907.03	-907.03	99,047.62
8	104,000	4952.38	99,047.62	0

6.9

Problem: Callable Corporation is offering a \$1,000 bond with semiannual coupons and a nominal coupon rate of 8% convertible semiannually. It is a five-year bond with call dates at the end of years 2 and 3. If the redemption value at each of these dates and also at maturity is \$1,060 and the price is \$1,022, find the possible yields to the investor and note which is least.

Solution. $F = 1000$, $m = 2$, $\alpha = 8\%$, $N = 5$

$$n = 10, r = 4\%, \text{ coupon} = 40, P = 1022, C = 1060$$

call date: EOY 2

$$1022 = 40 a_{\overline{4}|j} + 1060 (1+j)^{-4} \quad \leftarrow \text{biggest yield}$$

$$j = 4.77939503$$

call date: EOY 3

$$1022 = 40 a_{\overline{6}|j} + 1060 (1+j)^{-6}$$

$$j = 4.4679\%$$

maturity date: EOY 5

$$1022 = 40 a_{\overline{10}|j} + 1060 (1+j)^{-10} \quad \leftarrow \text{smallest yield}$$

$$j = 4.22043\%$$

6.9

Problem: A \$2,000 six-year 7% bond with annual coupons is purchased for \$2,200. If held to maturity, it is redeemable at par. The issuer may redeem the bond at the end of any year once it has been held for at least three years. The call premiums are \$150 at $t = 3$ years, \$140 at $t = 4$ years, and \$130 at $t = 5$ years. Determine the possible yields to the investor and note which is minimal and which is maximal.

Solution. $F = 2000, N = 6, \alpha = 7\%, m = 1$

$$n = 6, r = 7\%, P = 2200$$

If called at $k = 6 \rightarrow C = 2000$

If called at $k = 3 \rightarrow C = 2150$

If called at $k = 4 \rightarrow C = 2140$

If called at $k = 5 \rightarrow C = 2130$

$$2200 = 140 a_{\overline{6}|j} + 2000(1+j)^{-6} \Rightarrow j = 5.028057\%$$

$$2200 = 140 a_{\overline{5}|y_5} + 2130(1+y_5)^{-5} \Rightarrow y_5 = 5.796899\%$$

$$2200 = 140 a_{\overline{4}|y_4} + 2140(1+y_4)^{-4} \Rightarrow y_4 = 5.737774\%$$

$$2200 = 140 a_{\overline{3}|y_3} + 2150(1+y_3)^{-3} \Rightarrow y_3 = 5.64727\%$$

6.9

Problem: A \$1,000 10% ten-year bond with annual coupons is purchased for \$1,100. Its redemption amount at maturity is \$980 while it may be called in at the end of any year, the call premiums being \$115 at the ends of years 1–3, \$80 at the ends of years 4–6, and \$40 at the ends of years 7–9. Find the minimum yield rate to the purchaser.

Solution. $F=1000$, $\alpha=10\%$, $N=10$, $m=1$
 $n=10$, $r=10\%$. Coupon = 100, $P=1100$
 $C=980$

call at EOY 1–3 = $115 + 980 = 1095$
 call at EOY 4–6 = $80 + 980 = 1060$
 call at EOY 7–9 = $40 + 980 = 1020$

Min yield rate

EOY 1: 8.7272%

EOY 4: 8.287342%

EOY 7: 8.2828666% ← smallest

EOY 10: 8.350%